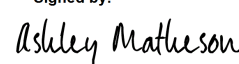
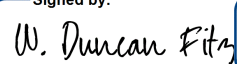
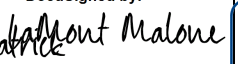
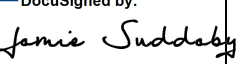




OIMS System 6-4 Work Management System

Comments:

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		SSHE – WMS STP-A Canada East	Hibernia Asset Manager – Pres HMDC	Hebron Asset Manager	Production Manager
Date	Reason for Issue	System Administrator- SME	System Administrator	System Administrator	System Owner

		Originated	Approved	Approved	Endorsed
Title: Work Management System- Appendix A					
Hebron Document Number: CAHE-EC-OOREF-01-006-4002-000					Rev: D14
Hibernia Document Number: HS-O-O-X-A00-XX-0604.007					
Revision Key: Typically D-Approved for Purchase, Construction or Implementation For details see OIMS System 4-1 Information Management.					

Record of Revisions

Revision	Section	Description
D14	All Sections	Integration of WMS 4.0 with PSMS, a revised Non-Permitted Activity List, approved single valve isolations for Hebron and Hibernia, removal of redundant information from the WMS manual, enhancements to the LTI Multi-Discipline Review Process, and updates to local regulations
D13	Section 11	Added WMS Addendum and updated Single Valve Isolation Lists to align with local OH&S regulatory requirements
D13	Section 12	Added Revised CSE ATMOSPHERIC TESTING REFERENCE VALUES & RPE REQUIREMENTS provided by MOH/IH
D12	Sections 11 & 12	Revised, consolidated and replaced existing Gas Testing Reference Charts with new “GAS/NORM TESTING REFERENCE VALUES & RPE REQUIREMENTS FOR CSE & GENERAL WORK” provided by MOH/IH
D12	Section 2 Permit Endorsement	Revised PIC responsibilities during Daily Work Coordination Meeting to evaluate work being performed against critical safeguards &, higher risk activities. Further revised that WMS assessments are assigned for these activities (addresses recommendation from OIMS assessment)
D12	Section 2 Authorization	Inserted table in Authorization section to define AA approvals for area and job activities (addresses OIMS assessment action item)
D12	Section 2 Workflows	Revised section 2 old DILO workflows to new Work Management Lifecycle and work flows for Permits, Isolations, Temporary Defeats and Job Safety Analysis. Aligns with Permit Vision Software.
D12	All sections	Replaced reference to SCC (Supplementary Control Certificate) with Temporary Defeat to align with WMS Best Practice 2020 Revision and ePTWS Software (Permit Vision)Terminology
D12	4.0	Newly Developed Hoarding Checklist added to SWP list
D 12	Signature Page	Updated to reflect position changes and new System 6-4 Administrator as document signatories
D11	1.0	Add statement for CCR Operator to countersign all Permits where the AO is a Disciplined Trades Person such as an Instrument Tech, Mechanic, Electrician working in an LER etc.

D11	12.0	Add Confined Space Entry Class Determination Tables as per WMS Updates.
D11	2.0	Update definition of a Temporary Defeat as per WMS updates
D10	11.0	Add Quick Reference Guide – Gas/Norm Testing and Respiratory Protection Requirements
D10	10.0	Update list of Low-Risk Single Valve Isolations for Hebron Platform.
D10	4.0	Add reference to SWP for Pyrophoric Material Management
D10	All Sections	Remove all reference to Sable Project
D9	3.0	To include reference to safety directive and deviation for Lifeboat PFD(Fall Prevention Devices)
D9	2.0	Update Permit Dilo Work Site Verification : Make changes to conditions that must be adhered to when AGT is delegating Gas testing for short term/low risk activities to Permit Holder
D8	2.0	Update to Permit Dilo: Pre-Issue Controls on the Permit DILO for Initial Gas Test no longer accurate since Initial Gas Tests cannot be electronically added in Permit Vision. Changed in Dilo
D8	2.0	Update to JASR Dilo: JSA & JSAR DILO changed to match the new terminology (Create vs. Utilize)
D8	2.0	Update to TD Dilo: TDC DILO changed to TD to match the new terminology (TDC vs. TD)
D8	8.0	Non permitted work list adjusted to add weekly lifeboat PM's as stand-alone JSA only activity
D8	10.0	Additional Drilling SBV's added to Hebron list
D7	2.0	Pg 12 - Permit Process Details Table (Permit Review Section) Additional Text added re: Gas Testing
D7	2.0	Pg 14 - Permit Process Details Table (Permit Issue Section) Additional Text added re: Gas Testing
D7	2.0	Pg 16 - Permit Process Details Table (Work Site Verification Section) Additional Text added re: Gas Testing
		Pg – New amended Isolation Dilo flow chart added to reflect changes in test to Out of Service equipment.

D7	2.0	Previous text under Out of Service Equipment = MOC created with instructions (MOC mandatory after 6 months) Revised text under Out of Service Equipment = - Equipment that is in LTI for a period greater than 6 months should be evaluated using the MOC Process - If equipment is deemed “out of service” it should be evaluated to determine if it should be removed, replaced or “positively isolated” using the Company’s MOC process.
D7	3.0	Text added to include reference to CNLOPB and CNSOPB RQFs applicable to Hebron, Hibernia and Sable Assets
D7	10.0	Pg 41 – List of Low-Risk Single Valve Isolations included for Hebron and Hibernia
D6	Section 1.0	EM Rig Supervisor role added to AA position within the WMS roles by position table
D5	Section 9.0	Pg-41, Electrical section of Personal Isolations updated to include following text. Testing and troubleshooting versus trouble shooting Lighting Repairs versus changing lamps on lighting circuits Testing/troubleshooting of heat trace and associated equipment
D5	Section 2.0	Pg-18, Text changes to Isolation DILO - ICC requirement to be filed with Isolation Control Point DILO - (<u>Isolation in Progress</u> section - #4, 5, 6, 7, Note and <u>Completed</u> section #4)
D5	Section 2.0	Pg 19,20,21 – Text changes to Isolation Process Details re: ICC w/ ICP
D5	Section 6.0	Pg-29, Functional Locks graphics updated
D5	Section 5.0	Pg-28, Added text for mandatory IC procedure attachment as per WMS 2.0 update
D5	Section 5.0	Pg-28, Added Permit pack required document checklist table
D5	Section 2.0	Pg-15, Changed text for “PH or Work Team Change out” as per WMS 2.0 update

D5	Section 2.0	Pg – 21, Changed DILO as per WMS 2.0 update
D5	Section 2.0	Pg—10, Changed MSDS to SDS as per WMS 2.0 update
D5	Section 1.0	Pg-7, Removed “Tow, Installation & Commissioning” roles & responsibilities Added “AO role can be assigned to discipline trade personnel based on direction of PIC/AA” into AO role of Typical WMS Roles table
D5	Introduction	Changed Deviation process text as per WMS 2.0 update
D4	Cover	Hebron cover page and approval signatures
D3	Section 8.0	Non permitted work list updated
D3	Section 3.0	Added Nova Scotia regulations
D3	Section 7.0	Included list of Sable SWPs
D3	Section 6.0	Canada East functional lock descriptions
D3	Section 7.0	Updated text in LSA and added section Rotating Equipment LSA
D3	Section 8.0	Revision to Control to Work to text, added to text describing process for creating asset specific non-permitted work list.

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1.0 Introduction

This document, termed “*Work Management System–Appendix A*,” is used for work execution in accordance with the Global Work Management System.

The Appendix **must** be used as specified as a supplement to the work requirements listed in the Work Management System.

The Appendix **will** be reviewed annually during the System 6-4 annual review.

Variance or Deviations from the requirements in this System (Critical Expectation [CriticalE] or Expectations [E]) by a Business Unit should be carefully considered.

Deviations from CE and E requirements in this System must be handled as follows:

Deviations must be reviewed by a multidisciplinary team of individuals knowledgeable about the issues associated with the proposed deviation

Knowledgeable personnel on the Variance/Deviation multidisciplinary team should consist of:

- The developer of the Variance/Deviation.
- A technical process representative (e.g., process engineer for the applicable unit);
- An hourly operation representative (e.g., experienced operator in the applicable unit);
- A health and safety representative
- Deviations from requirements must follow the deviation process described in the UBP Principles Document (Reference: “UBP Principles” Section 3.0 Control and Sustainment)
- No deviations are allowed for prohibited activities in the SIMOPS Matrix (Reference: Section 9.0 Simultaneous Operations)
- Approval for deviations is only granted if other safeguards will be introduced that maintain the safety of personnel, the environment, and facilities.

Example:

If physical disconnect or blinding for confined space entry cannot be achieved, then a deviation is required.

NOTE: This deviation would only apply to the specific job scope and duration

2.0 Permit to Work Overview

2.1 Purpose

- The purpose of Permit to Work System (PTWS) is to accomplish the following:
- Control the work activities and functional interfaces on Company facilities to prevent incidents resulting from uncontrolled work.
- Communicate information clearly to avoid an inadequate understanding of work task hazards.
- Gain necessary approvals and authorizations to execute work.
- Manage and control:
 - Risks relating to individual operations and work tasks
 - Procedures relating to the safety of individuals and facilities

2.2 Scope

The PTWS addresses the following work activities:

- **Hot Work**
- **Confined Space Entry**
- **Critical Work**
- **Energy Isolation**
- **Temporary Defeats of Critical Devices**
- **Atmospheric Testing**
- **Simultaneous Operations**

2.3 Overview

Several independent parties may be involved in controlling, coordinating, and managing activities. The following parties work together during the planning and execution phases of work:

Operations: is responsible for the care, custody, and control of the equipment and facilities.

- Work Team performs the work on the equipment and facilities
- **Site Management:** has oversight responsibility for PTWS to confirm that it is properly executed to confirm safe and effective work.

2.3.1 Operations

Operations are the owner of the facilities and have the expertise and skills to operate Company facilities. Operations have the following duties and expectations:

- Maintain monitoring and control of the operable systems and areas.
- Manage all required energy isolations.
- Safely prepare equipment and surrounding area prior to work commencement.
- Reinstate systems and isolations after completion of work.
- Maintain control of interfacing activities related to the operable systems and areas.

2.3.2 Work Team

- The Work Team may be a single worker or a group of workers. In either case, the person responsible for safely performing the work is the Permit Holder (PH). The PH confirms that personnel in the team possess the correct expertise and skills for performing the work activity.

The Work Team:

- Performs required safeguards related to the work activity.
- Confirms that work preparations and appropriate materials, tools, and equipment are in place.
- Confirms that the work site is safe in cooperation with the Area Operator (AO).
- Confirms that the work is executed according to the requirements of the Permit Pack.
- Recognize any changes in work scope that require work to STOP and changes to be addressed

2.3.3 Site Management

- Site Management is responsible for overseeing the execution of PTWS and has the following duties and expectations:
- Coordinates and supervises work on Company facilities
- Reviews the proposed work activity to confirm that potential interfaces are identified and appropriately managed.
- Review of work permits for completeness of job and safety planning efforts and endorsement of work permits as appropriate
- Assesses and stewards the effectiveness of PTWS in controlling work and in achieving the goal of safe and efficient work on Company facilities

2.4 Work Management Process

Hebron's Work Management system is an electronic system. Both paper and electronic permits can be issued for use at the work site. The system is designed to ensure the safe and efficient execution of work activities and integrates with the Personnel Safety Management System (PSMS) to enhance safety measures and operational efficiency.

Work Management System Best Practice - CAHE-EC-OOREF-01-006-4001-000 and the Work Management System - Appendix A - CAHE-EC-OOREF-01-006-4002-000 are the key documents supporting System 6-4. However, the Work Management Best Practice section 1.2 does state that all affiliates must also apply local regulatory requirements over and above the Best Practice requirements. Hebron will follow all OHS and Framework permitting requirements as outlined in sections under paragraph 4(1)(z), 101 and 102 of the Framework Regulations and Part 10 of the OHS Regulations.

The work permits are approved by a person other than the one who issues it and includes the following information:

- Name of the person who issued it (Permit Holder) and the person who approved it (Area Operator)
- The name of each person to whom it is issued
- The periods during which it is valid
- The work or activity to which it relates, including the location and any conditions the work/activity is subject to

Any safety or environmental risk and mitigations are identified (including physical and environmental conditions, any impediments to the proper use of any system or equipment and any simultaneous operations and reference to those activities, when applicable) and everyone involved in the permit will sign onto the permit and these signatures are attached to the work pack demonstrating that they read and understand its contents.

Under our Work Management process, permits will be used for the following activities:

- Hot Work
 - Confined space entry
 - Critical work
 - Testing/maintenance of the Fire & Gas System or Emergency Shutdown System)
 - Energy Isolation
 - Temporary defeats of critical devices
 - Working at Heights
 - Higher Risk Lifts
-
- **Detailed procedures for controlling specific hazards** are covered in separate documents which support the above process. Hebron's Work Management process guides personnel on or in the vicinity of the platform in performing tasks safely, consistently, efficiently, and in an environmentally responsible manner. The Work Management process also provides accurate information related to platform work activities during unplanned platform status changes. The use of the Work Management process is mandatory on the platform.
-
- **The Person in Charge (PIC)** has overall accountability for the execution of the Control of Work System at the field level. This accountability includes ensuring that trained and competent personnel are assigned to plan, approve, and execute work at the facility. Work Management process training and assessments are provided for personnel
-
- **The policy governing Work Management** ensures that all tasks are planned, coordinated, and carried out in a safe and environmentally responsible manner. Hazards to the personnel directly involved, other personnel, and the installation itself are identified, and measures are taken before work is carried out to ensure regulatory compliance and safe work practice
-
- **The Work Management process** is a risk-based system that requires a Permit to Work (PTW) be issued before a work task can be performed. While planning a PTW, the type of permit required must be identified; permit types include general, critical work, hot work, and confined space entry. The type of permit will determine how the risk will be managed and determines the level of authorization and controls needed to allow the work to be performed as safely as practical

3.0 PSMS Key Components

Pre-Task Briefing (PTB)

- Is a discussion led by the Supervisor (or trained designee) with the work team to focus the team on items important to safe work execution

Job Safety Analysis (JSA)

- Is a job-aid intended to identify task and site-specific hazards and safeguards necessary to prevent life-altering injuries or significant process safety events

Verification & Validation (V&V)

- A proactive field engagement process used by Supervisors to ensure life altering work safeguards are in place and effective.
- **Verification:** process of ensuring safeguards are in place (present)
- **Validation:** ensures the safeguards are effective (working as intended)

Life Saving Rules & Actions (LSRAs)

- LSRAs provide workers with a specific set of actions they must do to validate that safeguards are in place and effective prior to the start of work

Start Work Checks (SWC)

- Are used by the work team to verify that LSRA safeguards are in place and effective before and during the work/task

Hold Points / Triggers / Error-Prone Situations

- Human Performance concepts

START / STOP Work Expectations

- We will only START the job when safe to do so with effective safeguards in place.
- We will STOP the job if uncertain and / or unsafe to continue.

Last Minute Risk Assessment (LMRA)

- A Last-Minute Risk Assessment is a proactive, mental risk analysis tool for the individual. It is expected that individuals will use the card when they conduct the Last-Minute Risk Assessment to facilitate slow thinking

Targeted Learning Observation (TLO)

- Is a proactive learning process used to observe how work actually gets done compared to the job standard.

After Action Reviews (AAR)

- Is a quick and simple learning process that allows engagement between the leader and the team. It is facilitated by the supervisor with the people doing the work and will be conducted the majority of time to learn from successful work.

3.1 Work Management & PSMS Guiding Principles

PTW Integration

- A PTW is an integral part of the Work Management process and is required before performing any modification, maintenance, or breakdown work tasks on the platform

Risk Review

- All planned and unplanned work activities will be reviewed before the work starts to determine the level of risk associated with performing the work. The type of activity will determine the route each task takes throughout the Work Management process

Roles and Responsibilities

- Roles, responsibilities, and accountabilities of personnel involved with or affected by a work task will be clearly established

Communication

- The status of the ongoing task, from start to finish, will be communicated to all personnel involved with or affected by the work

PSMS Integration

- Integration with PSMS components such as Life Saving Rules & Actions (LSRAs) and Start Work Checks. Emphasis on risk assessment and management at the work face

Hazard Identification and Risk Management

- Based on these principles, the Work Management process encourages hazard identification, risk assessment, and risk management decision making to be carried out as close to the work face as possible. It requires that each task be documented, including hazard identification, risk assessment and risk management. This documentation may be audited and used to measure performance against accepted standards, as well as for continuous improvement

3.2 PTW Documentation

- The PTW is a formal written statement of scope, timing, location, hazards, mitigative measures, and the names and roles of personnel responsible and accountable. Executed certificates and PTW are stored on the platform at least 60 days, after which they will be transferred to the onshore Information Management Centre. There, they are inspected for quality and retained at least three years from the date of permit closure before being destroyed. Permit Holder, Area Authority, Area Interface Authority and Person in Charge, Isolating Authority, and Area Operator complete varying levels of training appropriate to their levels of responsibility within the Work Management process.

Examples for responsibilities of PTW are listed below:

3.2.1 Person in Charge (PIC)

- The PIC has primary responsibility for:
- Ensuring that competent personnel are assigned to plan, review, authorize and execute work at their facilities.
- Endorsement of work planning activities, including simultaneous operations (SIMOPS).
- Conducting Daily Work Coordination meetings and Endorse permits.
- Ensuring that trained and competent personnel are assigned to plan, approve, and execute work at their facilities.
- Oversight and compliance to the WMS.
- Assigning personnel to conduct V&V (verification and validation) activities (consistent with PSMS) and WMS assessments as well as conducting periodic V&V / assessments to understand WMS status and effectiveness.

3.2.2 Permit Holder (PH)

- The PH has primary responsibility for:
- Preparing and/or reviewing the Permit Pack documents to confirm accurate work scope, location, identification of hazards, applicable LSRAs, life-saving safeguards, controls for work and completing / facilitating the JSA when required
- Interfacing with the Area Authority
- Take part in the Pre-Task Briefing(s), including Start Work Check discussion(s) and reviewing Permit Pack documentation with the work team
- Ensuring START work readiness and reinforcing STOP Work Expectations with the team/individuals, and promote usage of LMRA
- Verifying and validating identified life-saving safeguards and work site controls are in place and effective
- Verifying and securing (lock and tag) the energy Isolation Control Point (ICP) (if applicable)
- Witnessing the demonstration of zero energy by the Isolating Authority before accepting the permit
- Completing work and restoring the work site to a clean and hazard-free condition
- Lead After Action Reviews (AARs) when planned / warranted

3.2.3 Area Authority (AA)

The AA has primary responsibility for:

- Reviewing Permit Packs for quality and completeness (e.g., job scope details, identification of LSRAs and life-saving safeguards, required SWP Checklists, JSA content (where required), procedures, Temporary Defeats).
- Ensuring competent individuals are assigned to execute tasks and appropriate tools are available.
- Verify isolation plans for their functional area of expertise (e.g. electrical, process, etc.)
- Authorizing permits to allow PH to commence work activity; withhold authorization until satisfied on readiness for safe execution
- Conducting periodic WMS V&V activities and assessments, reviewing results and validating lessons learned.

3.2.4 Area Operator (AO)

The AO has primary responsibility for:

- The operating equipment and systems in their area.
- The field review and issuance of the work permit to the PH.
- Conducting or assigning isolation and temporary defeat requirements.
- Acting as second verifier for isolations (if required).
- Reviewing and communicating work site controls and safe working processes to the PH.
- Verifying work status and restoring the equipment or system.
- Identifying how Zero Energy will be demonstrated to the Permit Holder.
- Demonstrating Zero Energy.

3.2.5 Isolating Authority (IA)

The IA has primary responsibility for:

- Identifying isolation requirements in association with the work scope and the PH.
- Applying all isolations in accordance with the WMS and safe work practices.
- Conducting and verifying isolations to ensure Zero Energy state is achieved and maintained.
- When requested, assist AO to demonstrate Zero Energy state to the PH.
- Removing isolations and restoring equipment and systems to normal

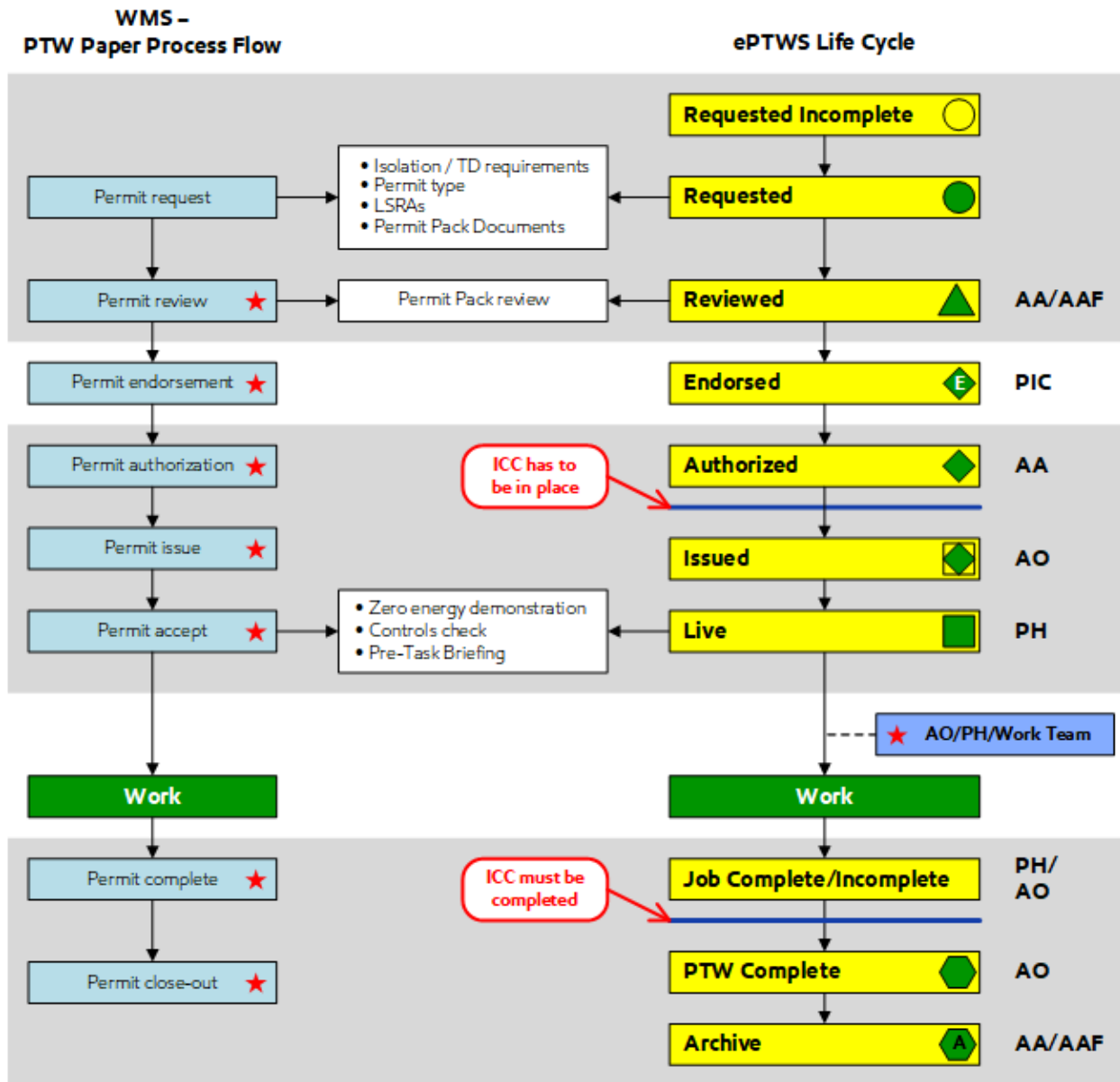
3.2.6 Site Coordinator (SC) / Area Interface Authority (AIA)

The SC/AIA is required (where applicable) to:

- Is designated as responsible for the work location where above activities are taking place
- Is typically the person responsible for the highest risk work at the location
- Understands what the higher risk activities are for the location which could lead to life-altering injuries or process safety events.
- Ensures work authorization process are followed
- Evaluates the proposed permit activities, location, and hazard controls, in relation to their existing work to verify awareness and provide endorsement for the work to proceed
- Identifies any barriers, scope changes, or competing priorities impacting ability to ensure work is executed safely (e.g., conflicting work interfaces, SIMOPS conflicts, significant scope changes).
- Notifies AA/AAF as needed and align on appropriate mitigations before proceeding
- Countersigns or signs the Permit to Work when managing activities that require coordination or interface with other work teams.
- May be a PH for one of the activities at site
- Has ultimate Stop Work Authority if work conflicts arise

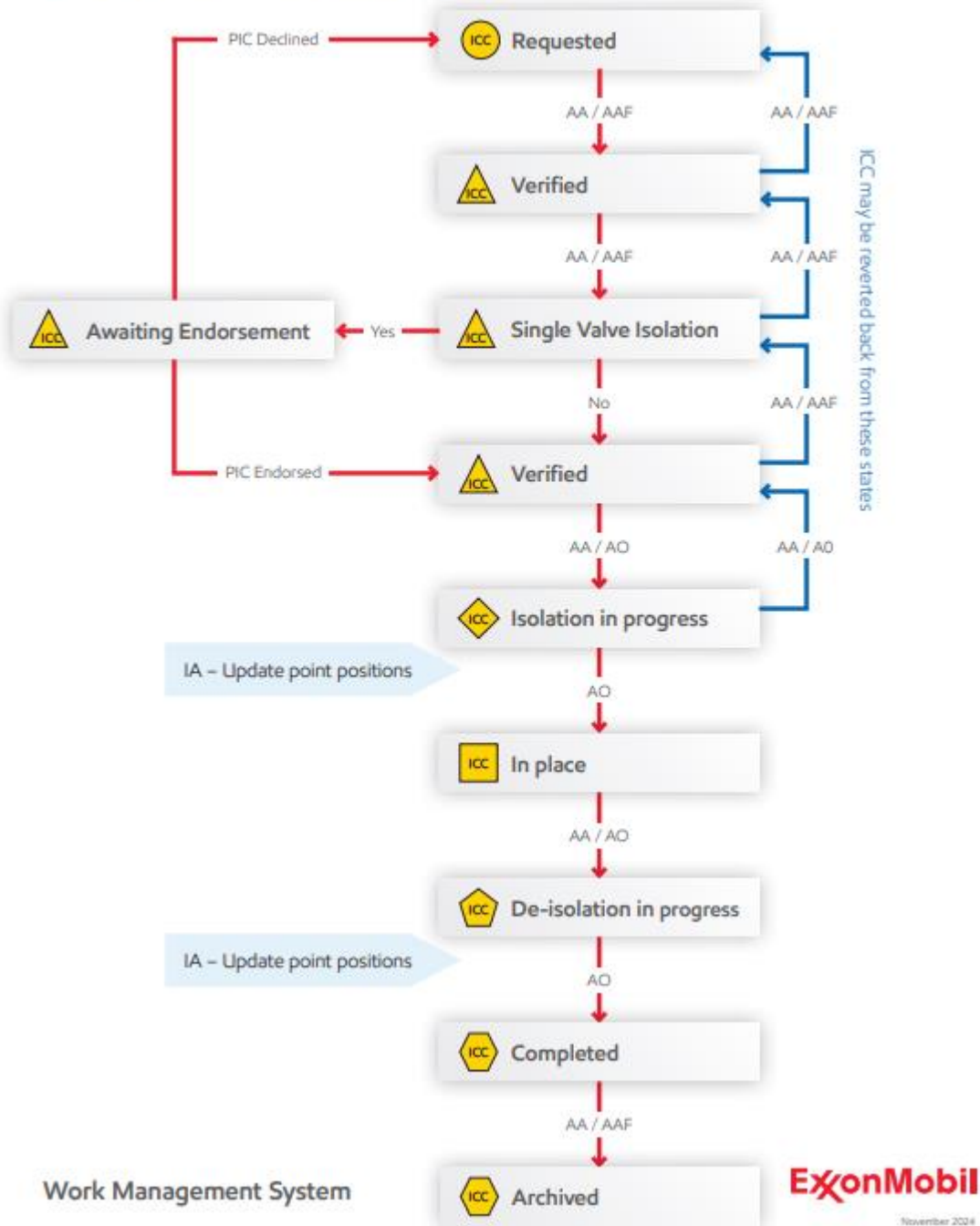
4.0 PTW Lifecycle

Permit to Work Process



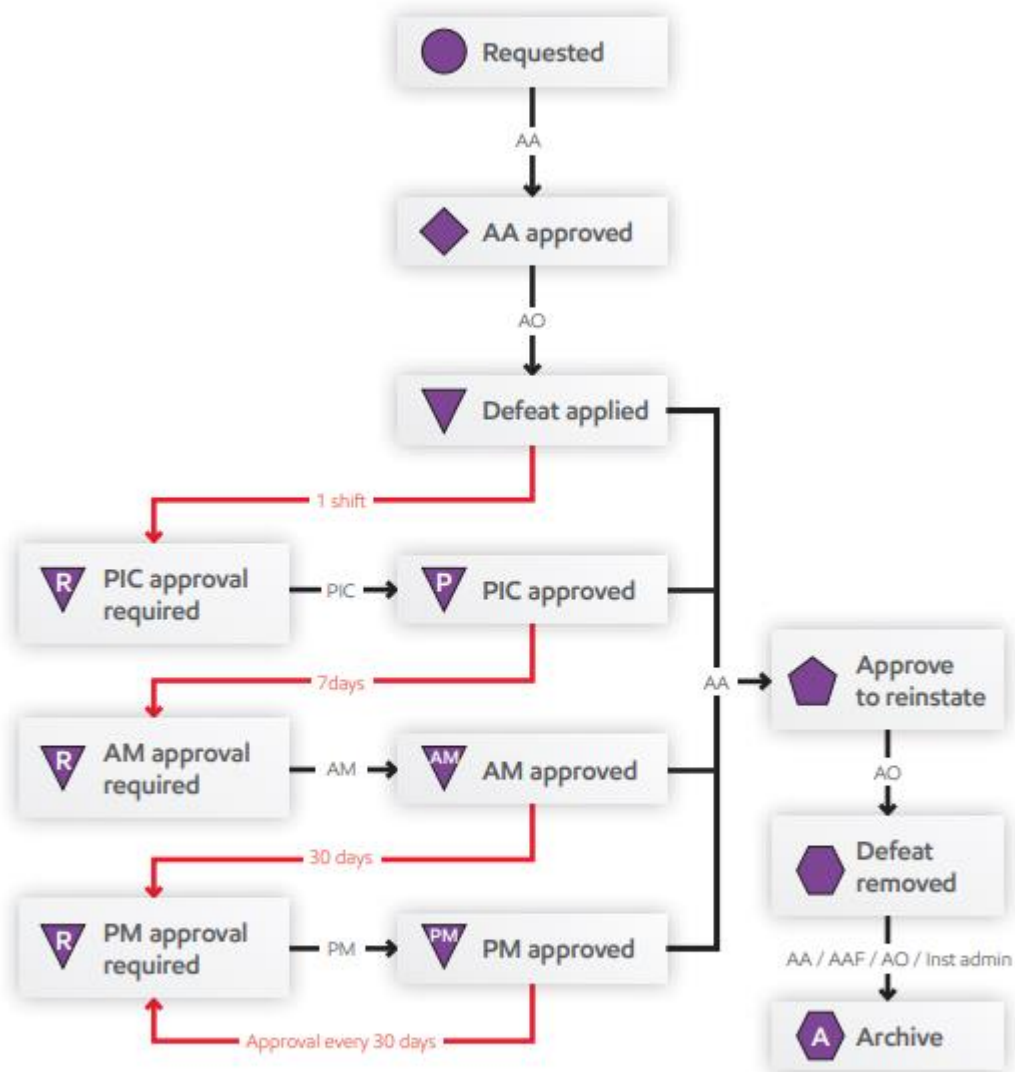
4.1 Isolation Lifecycle

Isolation Lifecycle



4.2 TD Lifecycle

Temporary Defeat Lifecycle

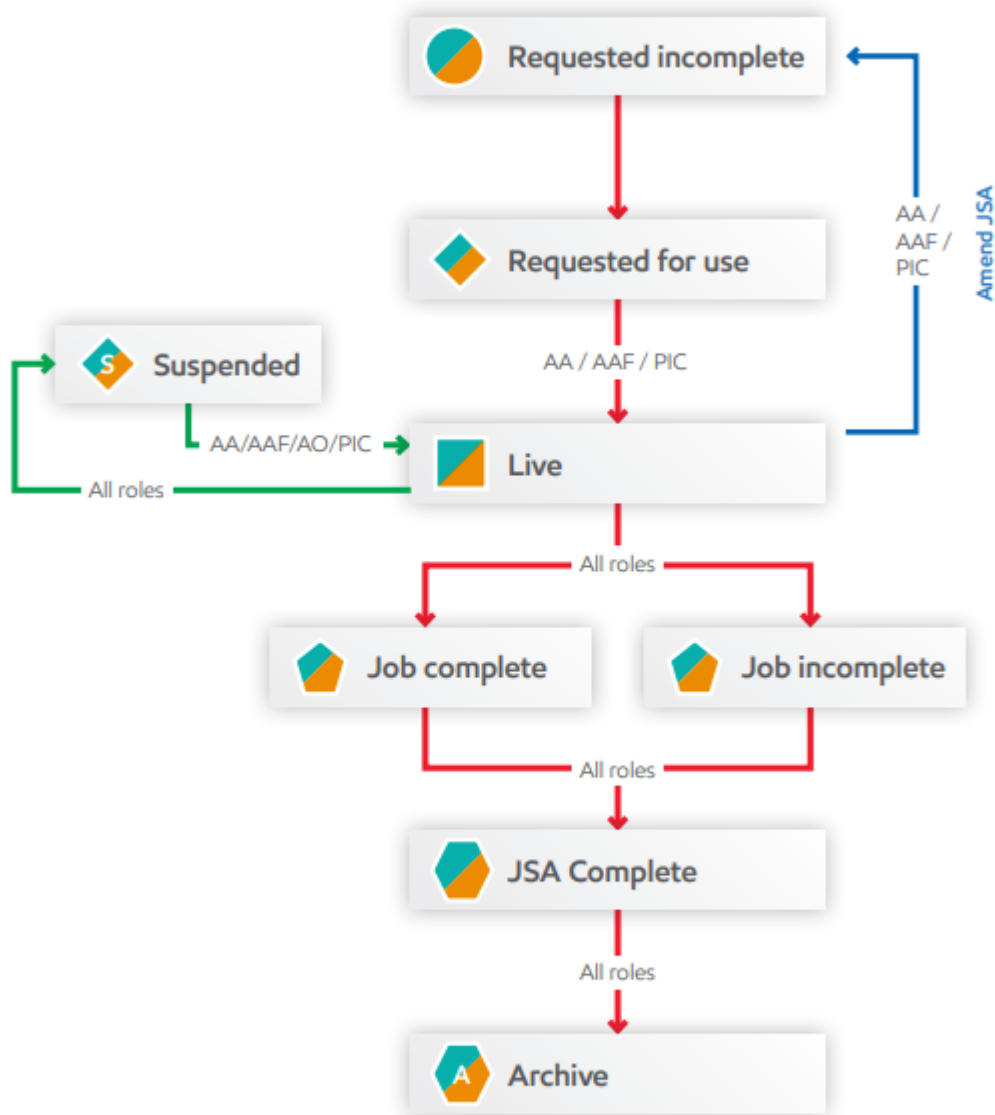


Work Management System

ExxonMobil
November 2024

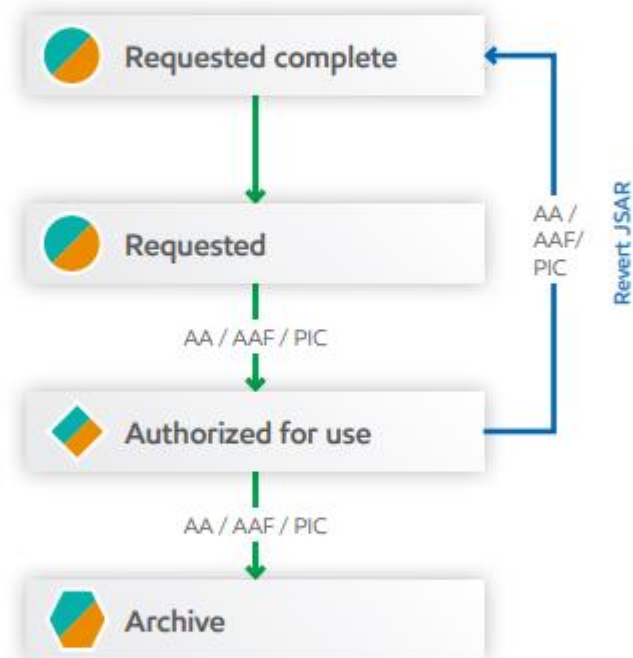
4.3 JSA Lifecycle

JSA Lifecycle



4.4 JSAR

JSAR Lifecycle



5.0 Local Regulatory Requirements

5.1 Laws and Regulations

Canada East will comply with all applicable laws and regulations, as well as adhere to all CNLOPB RQFs involving Hebron and Hibernia when applying this System including application of permissible mechanisms wherein the Company is granted written approval to operate under specific design codes, specifications, or protocols. In addition to the regulatory requirements, the Company may also place additional requirements on operations that will meet or exceed regulatory requirements.

These and other applicable subordinate regulations, Company-specific approvals, plus obligations deriving from them will be maintained in accordance with OIMS System 4-2 Compliance with Laws, Regulations, and Permits along with the Production Unit's Regulatory Compliance Manual.

References:

Canada–Newfoundland and Labrador Offshore Area Petroleum Operations Framework Regulations and OHS Regulations. All regulatory requirements, including RQFs, will be maintained in accordance with the latest CNLOPB Framework Regulations and the installation's Safety Management System (SMS).

5.2 Lifeboat Safety Directive

Due to several Marine Industry Fatal Lifeboat Incidents the following Executive Safety Directive has been issued:

Personnel will only be permitted to enter lifeboats during the following conditions:

- During emergency evacuation.
- During Maintenance Activities with Harbor Pins and Fall Preventer Devices (FPD) installed.

- Due to the design of the lifeboats and facilities on Hibernia and Hebron, personnel are required to enter lifeboats to install Fall Preventer Devices for a short duration (approx. 10 minutes) prior to completing the required maintenance, inspection and/or training
- This deviation from the Lifeboat Safety Directive has been risked assessed and fully approved in Risk Assessment-069-Personnel Entering Lifeboats without Fall Preventer Devices (FPD's) completed in October 1/2019
- To reduce any risks involved with the installation of the FDP's all maintenance activities, safety checks, inspections, etc. should be planned and scheduled to coincide with the installation of these devices during Sunday's Lifeboat Maintenance PM's
- Any unplanned task not requiring pendants would require a permit and would need PIC and AA approvals to conduct the work
- Lifeboat drills must be conducted at least every six months, including simulated or actual lifeboat boarding

5.3 Hot Work Safe Distances

- Hebron and Hibernia will follow CSA 117.2 **safe distance of 15m** as per section 140 (2) of the Canada–Newfoundland and Labrador Offshore Area Occupational Health and Safety Regulations.
- Hot work must be conducted under a permit system, including pre-work atmospheric testing, continuous gas monitoring in hazardous areas, and fire watch personnel.
- Firewater system readiness must be verified before hot work begins.
- Hot work is prohibited near fuel vents or intakes unless isolation measures are in place.

6.0 Items that Require a Permit (OSH Regulations)

6.1 Fall-Arrest and Travel Restraint Systems

- Any activity at a workplace that requires the use of a fall-arrest system or travel restraint system shall comply with Section 111 Require a Permit
- Fall protection equipment must be inspected before each use, and workers must be trained and certified in accordance with OHS Regulations

6.1.1 Lifts Using Materials Handling Equipment

- All lifts carried out at a workplace using materials handling equipment, except those classified under the occupational health and safety program as low risk

6.1.2 Occupation of a Confined Space

- Any occupation of a confined space at a workplace
- The method of entering and exiting the confined space;
- The escape route from the confined space, which, if persons are required to use a self-contained breathing apparatus or airline respirator, must allow for escape before the air supply is expected to be exhausted; and
- A record of all persons signing in and out of the confined space.
- A work permit for the occupation of a confined space ceases to be valid 12 hours after the most recent testing conducted.
- Continuous atmospheric monitoring is required in confined spaces where there is a potential for hazardous gas buildup.
- A standby attendant must remain at the entry point and maintain constant communication with the workers inside.

6.1.3 Hot Work

- All hot work carried out at a workplace on an installation must comply with Section 115(8)
- The location where the hot work is to be carried out, in particular, relative to Any areas referred to in subsection 26(2);
- The presence of any flammable, explosive, or combustible material; and
- The presence of any material that could produce toxic or flammable vapours.
- Hot work permits must specify safe distances from hazardous areas and require a fire risk assessment.
- A fire watch must be posted during and after hot work activities.

6.1.4 Hazardous Energy Exposure

- All work at a workplace that presents a risk of exposing any person to hazardous energy, including any work carried out closer to an exposed energized electrical conductor or circuit part than the applicable limited approach boundary or restricted approach boundary.
- A competent person must verify de-energization before work begins, and the isolation must be tested to confirm a zero-energy state.
- Locks and tags must remain in place until all work is completed, and no person other than the authorized worker may remove them

6.1.5 Activities Involving Explosives

- Any activity involving an explosive that is carried out at a workplace requires a permit

6.1.6 Work Permit Signatures

- The work permit must be signed by the person who issued it, the person who approved it, and every person involved in the activity to which it relates, to certify that they have read and understood its contents

7.0 Items that Require a Work Permit (Framework Regulations)

7.1 Hot Work on an Installation

- All hot work carried out on an installation.
- Additional details for the work permit – Expected to include:
- The work permit for hot work must set out safe distances to be maintained between the hot work and any well or any flammable, combustible, or explosive substance

7.1.1 Overriding the Fire and Gas Detection System

- Any work that involves overriding the fire and gas detection system shall require a permit
- Fire and gas detection system overrides must be as limited in scope and duration as possible
- Active inhibits on fire and gas detection systems must be logged and displayed in the control room for awareness

7.1.2 Testing and Maintenance of the Emergency Shutdown System

- Testing and maintenance of the emergency shutdown system must be conducted under a work permit
- No ESD override can remain in place longer than necessary, and all overrides must be removed immediately after testing
- Periodic full-scale ESD tests must be conducted in accordance with manufacturer and CNLOPB requirements.

8.0 Fire Water Isolation – Performance Standard Hebron and Hibernia

- Isolation valves shall be arranged and available such that no more than four (4) secondary firewater users (one section of ringmain) are isolated at any time.
- The requirement for no more than one section of the firewater distribution system (ring main) to be isolated is incorporated into the Permit-to-Work system.
- Firewater isolation should be planned during low-risk operational periods and must include alternative fire protection measures (e.g., portable extinguishers, dedicated fire watch)

9.0 Safe Work Practices

Hebron and Hibernia OIMS 6-4 Local Safe Work Practices and checklists

9.1 References

In addition to the ExxonMobil Work Management System, the following is a list of Hebron and Hibernia OIMS 6-4 Local Safe Work Practices and checklists:

- Abrasive Blasting
- Management of Temporary Equipment Offshore
- Guidance for use of Utility hoses and Tools
- Joint Control Process
- Welding Burning and Hot Cutting Operations
- Electrical Safe Work Practices
- Hoarding Checklist
- Pyrophoric Material Management

10.0 Life Saving Actions (LSRA)

12 Life Saving Rules & Actions

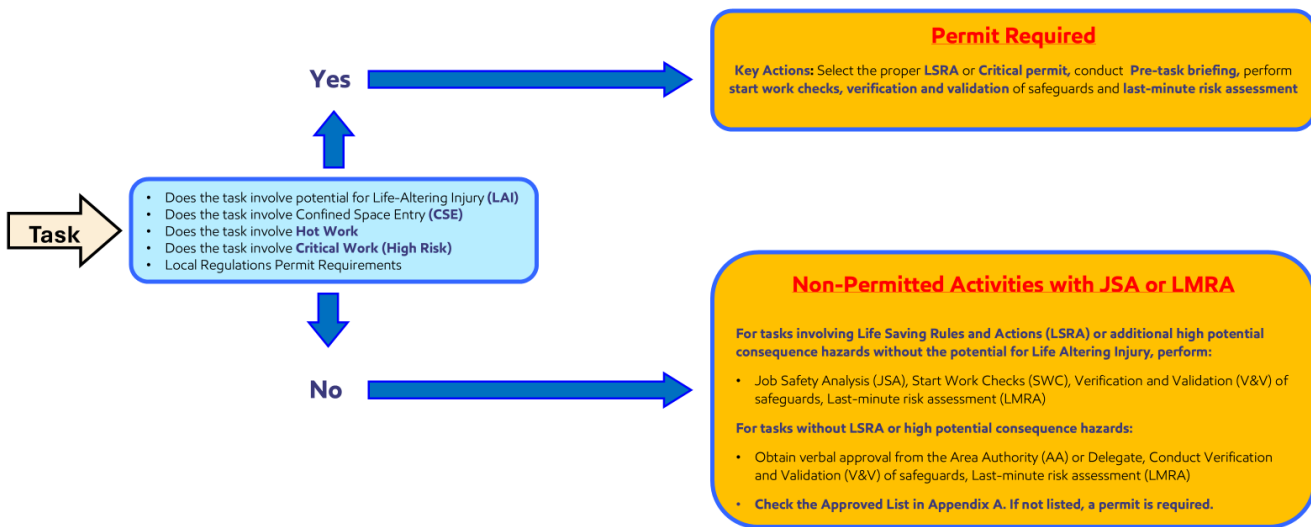
The Company has LSRAs for work activities involving higher potential consequences. The LSRAs provide workers with a specific set of actions they must do to validate that safeguards are in place and effective prior to the start of work:

Working at Height Protect yourself against a fall when working at height  1. I inspect my fall protection equipment before use 2. I secure tools and work materials to prevent dropped objects 3. I tie off 100% to approved anchor points while outside a protected area 4. I have validated prompt rescue capability when wearing fall protection 5. I will follow safe work practices when climbing, descending or working from ladders	Energy Isolation Verify isolation and zero energy before work begins  1. I have identified all energy sources 2. I confirm that hazardous energy sources have been isolated, locked, and tagged 3. I have checked there is zero energy and tested for residual or stored energy 4. I never remove or tamper with energy control devices without authorization 5. I confirm equipment is back to safe operating condition prior to re-energizing	Bypassing Safety Controls Obtain authorization before overriding or disabling safety controls  1. I understand and use safety-critical equipment and procedures which apply to my task 2. I obtain authorization before: - Disabling or overriding safety equipment - Deviating from procedures - Crossing a barrier	Line of Fire Keep yourself and others out of the line of fire  1. I position myself to avoid: - Moving objects - Vehicles - Pressure releases - Dropped objects 2. I establish and obey barriers and exclusion zones 3. I take action to secure loose objects and report potential dropped objects
Confined Space Entry Obtain authorization before entering a confined space  1. I confirm energy sources are isolated 2. I confirm the atmosphere has been tested and is monitored 3. I check and use my breathing apparatus when required 4. I confirm there is an attendant standing by 5. I confirm a rescue plan is in place 6. I obtain authorization to enter	Work Authorization Work with a valid permit when required  1. I have confirmed if a permit is required 2. I am authorized to perform the work 3. I understand the permit 4. I have confirmed that hazards are controlled and it is safe to start 5. I stop and reassess if conditions change	Driving Follow safe driving rules  1. I always wear a seatbelt 2. I do not exceed the speed limit, and reduce my speed for road conditions 3. I do not use phones or operate devices while driving 4. I am fit, rested and fully alert while driving 5. I follow journey management requirements	Safe Mechanical Lifting Plan lifting operations and control the area  1. I confirm that the equipment and load have been inspected and are fit for purpose 2. I only operate equipment that I am qualified to use 3. I establish and obey barriers and exclusion zones 4. I never walk under a suspended load 5. I understand lift plan prior to starting work and keep hands off load unless authorized 6. I establish and maintain communication with lifting and rigging personnel
Excavations Obtain authorization before digging or entering excavations  1. I obtain authorization prior to performing excavation / trenching / ground disturbance activities 2. I confirm underground utilities and structures are located, de-energized or protected 3. I confirm hazard controls and confined space entry as required 4. I monitor for changing conditions and exit as appropriate	Hot Work Control flammables and ignition sources  1. I identify and control ignition sources 2. Before starting any hot work: - I confirm flammable material has been removed or isolated - I obtain authorization 3. Before starting hot work in a hazardous area I confirm: - a gas test has been completed - gas will be monitored continually	Breaking Containment / OPE Manage ignition sources and exposure to energy & hazardous substances  1. I verify opening/break location(s) 2. I confirm energy isolation is complete 3. I manage ignition sources and confirm gas testing is completed when required 4. I use appropriate PPE and defensive behaviors when opening process equipment	Work Around Mobile Equipment See and be seen; maintain a safe distance  1. I plan work to separate workers and mobile equipment 2. I only operate mobile equipment if qualified and use spotters where required 3. I verify safety devices are functional on mobile equipment 4. I establish clear means of communication with equipment operator 5. I stay out of the equipment's exclusion zone while in operation 6. I confirm controls are in place to prevent unintentional movement of mobile equipment and/or loads

11.0 Work Management System Flowchart

The Work Management System Flowchart will help identify when work is permitted or non-permitted, and which PSMS tools are required. Review the Non-Permitted Work list; if your task is not on the list, a permit is required. If you believe your task should be on the list, discuss it with your supervisor and forward it to the WMS advisor for review.

Work Management System Flowchart



11.1 Appendix A: Non-Permitted Electrical Chart

	Intrusive Maintenance/Repairs			Testing/Troubleshooting		
Voltage	Maintenance Activities Electrical Safe Work Condition	Maintenance Activities Proximity Work	Maintenance Activities Energized Equipment	Testing Electrical Safe Work Condition	Proximity Work	Energized Equipment
50V or less (Intrinsically Safe)	LMRA	LMRA	LMRA	LMRA	LMRA	LMRA
50V or less (non-intrinsically safe)	LMRA	LMRA Spark potential JSA: For unclassified areas	LMRA Spark potential JSA: For unclassified areas	LMRA Spark potential JSA: For unclassified areas	LMRA Spark potential JSA: For unclassified areas	LMRA Spark potential JSA: For unclassified areas
50V to 300V	Spark potential JSA	Spark potential JSA	Critical Permit: Standby person required.	LMRA Spark potential JSA: For unclassified areas	LMRA Spark potential JSA: For unclassified areas	Critical Permit: Standby person required.
300V to 750V	Spark potential JSA	HRE JSA: Attach atmospheric gas test if in unclassified area	Critical Permit HRE: Standby person required.	LMRA Spark potential JSA: For unclassified areas	LMRA Spark potential JSA: For unclassified areas	Critical Permit HRE: Required Standby person required
750V and above	Critical Permit HRE: Standby person required.	Critical Permit HRE: Standby person required.	Critical Permit HRE: Standby person required.	Critical Permit HRE: Standby person required.	Critical Permit HRE: Standby person required.	Critical Permit HRE: Required Standby person required.

11.2 Approved Non-Permitted Work List

Approved Non-Permitted Work List		
Any work activities not listed below should be evaluated for PTW Requirements  Any JSA work involving LSRAs requires SWC reviews Any work with LAI potential requires a PTB (*)Denotes: Spark Potential and Ignition Sources tasks require an Authorized Gas Tester (or designate), Atmospheric Test Record and Permission From Operations prior to introducing the ignition source in the hydrocarbon areas.		
Lead Discipline	Task	Control Process JSA or LMRA
General	Non-Intrinsically Safe Equipment: Includes devices such as cameras, laptops and Surveying Equipment	JSA*
	Low-risk work requiring Energy Isolation (e.g. Drilling equipment, Utility systems, Water leaks, Oil changes, non-hydrocarbon PSVs, Filters/strainers, Greasing Belt changes, Equipment checks, Low-risk elevator maintenance, laundry servicing, HVAC filters.	JSA/SWC
	Non-Complex Lifts: Use of Nylon Slings	JSA
	Pneumatic tools with Spark Potential (e.g. impact wrench, impact gun, etc.)	JSA*
	Material Movement and Equipment Repairs (Spark Potential): Includes tasks such as moving materials, using hand tools, torque checks, ponding grating, installing handrails, mesh screens, tray clashes, Installing or Re-locating Cable Tray, supports, addressing water leaks with clamps or wraps, installing/removing Trelleborg seal, maintaining doors and repairing galley equipment etc. No Isolation Requirements	JSA*

	Repair Shop Work: Excludes areas where an LSRA exists. Note: hot work requires a permit due to local regulations	LMRA
	Outdoor Housekeeping and Maintenance: Tasks include deck cleaning, aerosol and brush painting, bulb replacement, manual snow removal, and salt/sand treatment	LMRA
	Temporary power hook-ups (running and plugging / unplugging cords)	LMRA
	Housekeeping and Maintenance Within Accommodations/Office: Excluding where an LSRA exists	LMRA
	Erecting Rigging Equipment: (No WAH)	LMRA
	Pressurized Habitat Build & Operate	LMRA
	Bunkering Fluids: Follow established procedures	LMRA
	Material Movement and Equipment Repairs (No Spark Potential): Includes tasks such as moving materials, using hand tools, torque checks, ponding grating, installing handrails, mesh screens, tray clashes, Installing or Re-locating Cable Tray, supports, addressing water leaks with clamps or wraps, installing/removing Trelleborg seal, maintaining doors and repairing galley equipment etc. No Isolation Requirements	LMRA
	Pressure Washing: Plant pressure	LMRA
	Pneumatic tools without Spark Potential	LMRA
	Staging Tools and Equipment into Work Areas: Includes tasks such as material movement and mobilizing equipment to worksites	LMRA
Construction	Needle Gun, Bristle Blasting	JSA*
	Install and Removal of Cladding/Insulation with EHT	JSA
	Install and Removal of Cladding/Insulation without EHT	LMRA
	Pulling Instrument / Electrical Cables / Glanding Instrument / Electrical cables (No WAH)	LMRA
	Installing/Repairing: Flow coding, process tags, limacoid, signs, labels etc	LMRA
	Field Install of Non-Energized Equipment : Pre-tie-in (running tubing, tray, field instruments, EHT, cables)	LMRA

	Installing Techno Wrap (not including bristle blasting)	LMRA
	General Scaffold Build/Inspections: Below 6' (e.g., hoardings, step-ups, lifting frames)	LMRA
	Install and Removal of Insulation Bags	LMRA
	Roller/Brush Painting & Belzona Application	LMRA
Drilling	Routine Deck Operations - Low Risk (with no reasonable LAI potential) involving LSRA's	JSA/SWC
	Radioactive Source: Follow established procedures	JSA
	Non-Routine LSRA Drilling tasks executed using established procedures & practices (with no reasonable LAI potential)	JSA/SWC
	Non-Complex Lift Using Pipe Deck Gantry	JSA/PTB
	Work on Inactive Rig Circulating System (HP Mud Manifold, Trip Tank, C/K Manifold) When Not Connected to Mud Pumps (No LAI Potential) Hibernia.	JSA
	Slip and Cut Drill Line: LAI activities (e.g., man riding) covered under a separate WAH permit	JSA
	Routine Rig Floor Operations - Tasks include TIH, POOH, circulating, reaming, Tugger operations, drilling, etc.	LMRA
	Cleaning Trip Tank: Vented, cleaning from outside the hatch (No CSE)	LMRA
	Using Automated Internal Tank Cleaning System: Rotating sprinkler head	LMRA
	Routine Drilling Shaker Operations	LMRA
	CRI Ops: Bulk transfers, mixing product, pumping cement	LMRA
	Routine maintenance tasks performed by drilling personnel according to established procedures. For non-routine or invasive work (e.g., isolation, BC), a PTW will be utilized, ensuring no reasonable risk of Life Altering Injury (LAI)	LMRA
	Low Risk Tasks: Equipment setup (hoses, pumps), low pressure/non-hydrocarbon pressure testing/service testing, purging, flushing, draining, performing low risk PMs, CA testing, Low volume pressure or service testing	LMRA

Maintenance	Maintenance & Inspection: Includes tasks such as checking junction boxes and control panels.	JSA*
	Radioactive Source: Follow established procedures	JSA
	Low Risk Work with Isolation Requirements: Drilling equipment, utility systems, water leaks, oil changes, non-hydrocarbon PSVs, filters/strainers, greasing, belt changes, equipment checks, low risk tasks inside the elevator car or machinery maintenance (no CSE), laundry servicing, HVAC filters, hydraulic leak repairs, Galley equipment etc.	JSA/SWC
	Low Risk Intrusive IT Troubleshooting tasks outside of accommodations or electrical/instrument rooms	JSA*
	Inspection of Piping, External Tanks/Vessels, Structures, Concrete, and Equipment: Conducted using pulsed eddy current (PEC), magnetic particle inspection (MPI), ultrasonic testing (UT), forward-looking infrared (FLIR), liquid penetrant testing (LPI), and digital cameras for reporting in process areas. (Spark Potential)	JSA*
	General Visual Inspection/Close Visual Inspection (Spark Potential)	JSA*
	Calibrating and Adjusting Instruments (Intrusive): This includes transmitters, temperature/pressure gauges, flow meters, density devices, oil-in-water analyzers, storage silos, and well control systems, excluding Fire & Gas (F&G) and Emergency Shutdown (ESD) systems, as per local regulatory requirements. LSRA for Energy Isolation	JSA*/SWC
	PCS/PLC/PSD and DSD System Maintenance (Intrusive): Excluding F&G and ESD systems (Regulatory Requirement)	JSA*
	PAGA System Routine Maintenance (Intrusive): PMs, regular function testing, minor repairs on PAGA system components (e.g., beacons, horns)	JSA*
	Troubleshooting radios/emergency communications, CCTV, data cables, phones, HAEQ inspections, FDAP, WAPs, cable running, PAGA, weather systems, SART/EPIRB, GMDSS, driller's equipment (Intrusive)	JSA*
	Re-lamping of lights in classified areas.	JSA*

	Low Risk Non-intrusive IT Troubleshooting tasks	LMRA
	Inspection of Piping, External Tanks/Vessels, Structures, Concrete, and Equipment: Conducted using pulsed eddy current (PEC), magnetic particle inspection (MPI), ultrasonic testing (UT), forward-looking infrared (FLIR), liquid penetrant testing (LPI), and digital cameras for reporting in process areas. (No Spark Potential)	LMRA
	General Visual Inspection/Close Visual Inspection with IS equipment	LMRA
	Calibrating and Adjusting Instruments (Non-Intrusive): This includes transmitters, temperature/pressure gauges, flow meters, density devices, oil-in-water analyzers, storage silos, and well control systems, excluding Fire & Gas (F&G) and Emergency Shutdown (ESD) systems, as per local regulatory requirements. For intrusive tasks, isolation and verification of zero energy are required.	LMRA
	PCS/PLC/PSD and DSD System Maintenance (Non-intrusive): Excluding F&G and ESD systems (Regulatory)	LMRA
	Troubleshooting radios/emergency communications, CCTV, data cables, phones, HAEQ inspections, FDAP, WAPs, cable running, PAGA, weather systems, SART/EPIRB, GMDSS, driller's equipment (non-intrusive)	LMRA
	Operating Air Compressor: Filling SCBA using a non-intrinsically safe air compressor in a non-hazardous area.	LMRA
	Noise Monitoring	LMRA
	Cathodic System Maintenance: Measurement and monitoring activities outside process areas	LMRA
	Remove and Reinstall Clamp-on Flow Meters	LMRA
	Performing an oil change on non-intrusive equipment	LMRA
	1W/1M/3M Lifeboat PMs	LMRA
	Pulling, Glanding Cables	LMRA
	Re-lamping of lights in non-classified areas.	LMRA
	HAEQ, other routine visual inspections.	LMRA

	Perform TS&R or calibration checks on intrinsically safe transmitters	LMRA
	Black Start Compressor operations	LMRA
Operations	Decanting Chemicals & Sampling of Process and Utility Systems: Streams with H2S Potential > IDLH	JSA
	Decanting Chemicals & Sampling of Process and Utility Systems: Streams with H2S Potential < IDLH	LMRA
	Low Risk Tasks: Equipment setup (hoses, pumps), low pressure/non-hydrocarbon pressure testing/service testing, purging, flushing, draining, performing low risk PMs, CA testing, Low volume pressure or service testing	LMRA
	Chemical Dosing: Excluding Acrolein	LMRA
	Executing Equipment/System Operating Procedures: No LAI potential activity	LMRA
	Routine Production Operation: Includes operator rounds, resetting/restarting/stopping equipment, lubrication, checking oil levels, refilling, visual inspection, sampling and gas testing	LMRA
	ABB Support: Excluding F&G and ESD Systems	LMRA
	Equipment Walk Downs/Inspections	LMRA
	Installing/Removing Isolations, Draining, Purging, Flushing, Venting, Proving Zero Energy	LMRA
Services	Use of Non-IS Grease Gun	JSA*
	Weekly PMs and Inspections (e.g. fire extinguisher checks, lifesaving equipment, PPE, etc.)	LMRA
	Daily/Weekly/Monthly Crane Checks: No WAH	LMRA
	Use of VIPER Grease Gun	LMRA
	The use of Nylon Slings for normal lifts only	JSA
	Helicopter Operations: Includes landing, takeoff, and jet fueling	LMRA
Wellwork	Non-Routine LSRA Wellwork tasks executed using established procedures & practices (with no reasonable LAI potential)	JSA/SWC

	Non-Complex Lift Using BOP Crane: Includes low-risk lifts performed close to the deck (below knee level) and along the beams	JSA/PTB
	Laying out Treating Iron	LMRA
	Moving the A Frame: On-wheels, no complex lift involved	LMRA
	Routine low-risk Wellwork tasks executed using established procedures & practices (with no reasonable LAI potential)	LMRA

12.0 Tasks Performed Using Personal Isolation

12.1 Personal Isolation

Personal Isolations can be performed for lower risk, short duration tasks such as:

- Taking samples
- Isolating instruments
- Replacing pressure gauges
- Replacing small filter cartridges or small valves
- Small lighting circuit checks/lamp replacements

12.2 Control of Activity

The personal isolation must provide care, custody and control of the equipment to the person conducting the work (Permit Holder). Isolations will typically be visible to the PH (adjacent to the work face). If a point is not visible, the isolation points locked and tagged with a functional lock and key and will be retained by the PH.

12.3 Personal Isolation Criteria

The criteria to meet for personal isolation is as follows:

- Before approval the AA or designate must be satisfied with the competence of the person conducting the work to execute the isolation and to complete the task safely. The AA can request the AO to assist with the isolation if required.
- De-/Isolation takes place immediately after the task has been completed.
- An ICC is not required if the task is completed before the end of shift. An ICC must be processed if not completed by the end of the shift
- The person conducting the work may receive assistance from others, but the person must always be present while the work is in progress
- The person conducting the work must confirm zero energy and demonstrate zero energy to anyone assisting with the work.
- If the work site must be left unattended during the shift, the equipment will be isolated, locked and tagged. (Made Safe)
- During permit close out, the AO must confirm that the isolation points are returned to the normal state.

12.4 Approval Personal Isolations List

Operations	
	Taking normal / routine oil and water samples from designated (engineered) sampling points
	Stroking of valves for maintenance checks (no breaking containment)
	Changing suction filters
	Routine running of diesel generators
	Deluge testing (short duration)
Instrumentation	
	Instrumentation local isolations for transmitters, gauges, switches, troubleshooting, calibration & function testing.
	Repair of I.S. instruments / Troubleshoot LV circuits
	Remove /Repair Instrument tubing
Electrical	
	Testing/Troubleshooting of electrical equipment (LV circuits)
	Lighting Repairs
	Elevator troubleshooting / maintenance
	Testing/troubleshooting of heat trace and associated equipment
Mechanical	
	Maintenance / Routine trouble shooting of equipment, filter changes, hydraulic systems (non-intrusive), small pump oil changes.
	Greasing valves /machinery (non-intrusive)
	Fire main and hydrant work of short duration.
	Air operated overhead crane maintenance
	Air operated door maintenance
All Departments	
	HVAC - first line maintenance checks, Visual (Does not include equipment strip down or belt replacement)
	Minor repairs to galley equipment

Drilling	
	Replacing hoses or O-rings at deck level
	Inspect/adjust/grease draw works brakes.
	Troubleshoot PHM, PLS /EFM to obtain pressure readings or reset speed control or counterbalance cartridges.
	Troubleshoot VPH / IR /CWM / MB /Catheads RT /Mousehole

13.0 Long Term Isolation (LTI) Multi-Discipline Review Process

- The LTI Multi-Discipline Review Process ensures compliance with the WMS Manual through structured evaluations and risk management. The multi-discipline team meets monthly to review LTIs, maintaining an updated register to track status, repair schedules, and potential disconnection requirements. LTIs unresolved after six months are escalated to the Asset Manager for evaluation, prioritizing resolution, risk mitigation, or disconnection through the Management of Change (MOC) process if necessary.
- The OMT team is responsible for setting up these reviews with the Asset Manager. The OMT team and the Asset Manager meet every six months to review unresolved or critical LTIs.
- Periodic risk assessments address potential hazards like corrosion, dead legs, and automation loss, ensuring operational safety and compliance.

14.0 WMS Addendum – Single Valve Isolations

14.1 Purpose

This process has been developed to provide an overview of Hebron and Hibernia's Piping Isolation philosophy, specifically to address the additional isolation requirements of the Canada-NL Offshore Area Occupational Health and Safety Regulations.

Section 144(3) of the Regs state:

(3) The employer must ensure that

- an energy-isolating device used on a pipe that contains a substance that may release hazardous energy, other than in a confined space,
 - consists of a blank or blind in conjunction with valves or other blocking seals that are secured and locked out in the closed position to prevent the substance from reaching the blank or blind,
 - consists of a double block and bleed system consisting of two valves or other blocking seals that are secured and locked out in the closed position and located on each side of a valve or other mechanism
 - that is secured and locked out in the open position to allow for bleed-off between the two seals,
- or
- has been approved by a professional engineer;

14.2 Introduction to WMS Best Practices

Hibernia and Hebron utilize ExxonMobil's Global Work Management System (WMS) Best Practice and perform energy isolations in accordance with Minimum Isolation Standards and required practices of this global practice and as outlined in the Operations Integrity Management System 6-4 "Work Management". This Best Practice has been developed and implemented globally based on accepted industry practices and aligned with equipment and devices designed for facility application based on API and ExxonMobil Engineering Global Practices (GP) requirements and guidance.

14.3 Minimum Isolation Standards

Outlined in the Minimum Isolation standards are the use of Positive and Double Block and Bleed Isolation methods as described in Section 144 (3)(a)(i) and 144 (3)(a) (ii) of the OHS Regulations. The standards also include additional isolation methods where Positive Isolation or Double Block and Bleeds are not possible and other accepted methods given facility design or lower risk operating conditions of the system or equipment in service. All isolations that are planned or performed include required authorizations and approvals and are subject to a number of additional controls which may include Draining, Depressuring, Leak and Integrity Testing, Lock and Tag-Out Requirements, Independent Verification, and Zero Energy Demonstration).

14.4 Isolation Methods and Controls

In compliance with the new OHS Requirements, this addendum has been developed to obtain Professional Engineer approval of the Hebron and Hibernia WMS isolation methods, standards, and practices including a P. Engineer approved list of Isolations using Single Valve Isolation.

14.5 Isolation Methods

Isolation Method Standards			
Isolation Type	Definition	Illustrations	
Blinding Spading or Disconnection	Installation of a physical barrier (blind or spade) to separate energy from an activity Or Physical disconnect		
Double Isolation and Bleed	Two independent block valves in series with a bleed valve between them (i.e. 2 independent gate, globe, ball, plug, or needle valves)		

Isolation Method Standards			
Isolation Type	Definition	Illustrations	
Double Isolation and Bleed Valves	A single valve assembly with: 2 independent obturators in a single common valve body with a bleed between or Expanding gate or expanding plug valve with a single obturator with two mechanical seats that do not require pressure to seal (Refer to Type C Valve in GP 03-12-02U & 03-01-01U for use limitations)		
Single Valve Isolation	A Single Valve Isolation (SVI) consists of the closure of a single isolation valve.		

Table 14: Minimum Isolation Standard Table

Fluid	Flange Class	Min isolation standards	Conditions if Double Isolation Standard is Not Met
Flammable (e.g. crude oil, methanol, diluent, dilbit, emulsion, solvent and diluted froth) Includes combustibles above flash point or <8°C/15°F from flash point	≥ 300	Double isolation & bleed, or Blind and/or physical disconnect	Condition A
	< 300	Double isolation & bleed, or Blind and/or physical disconnect	Condition B
Toxic or harmful (e.g. H ₂ S, acids, caustics and amine)	All	Double isolation & bleed, or Blind and/or physical disconnect	Condition A
Non-flammable and operating at > 66°C /150°F (e.g. steam, boiler feed water)	≥ 300	Double isolation & bleed, or Blind and/or physical disconnect	Condition B
	< 300	Single isolation	N/A
Non-flammable and operating at ≤ 66°C/150°F (e.g. lube oil, water, glycol, air)	≥ 900	Double isolation & bleed, or Blind and/or physical disconnect	Condition B
	< 900	Single isolation	N/A

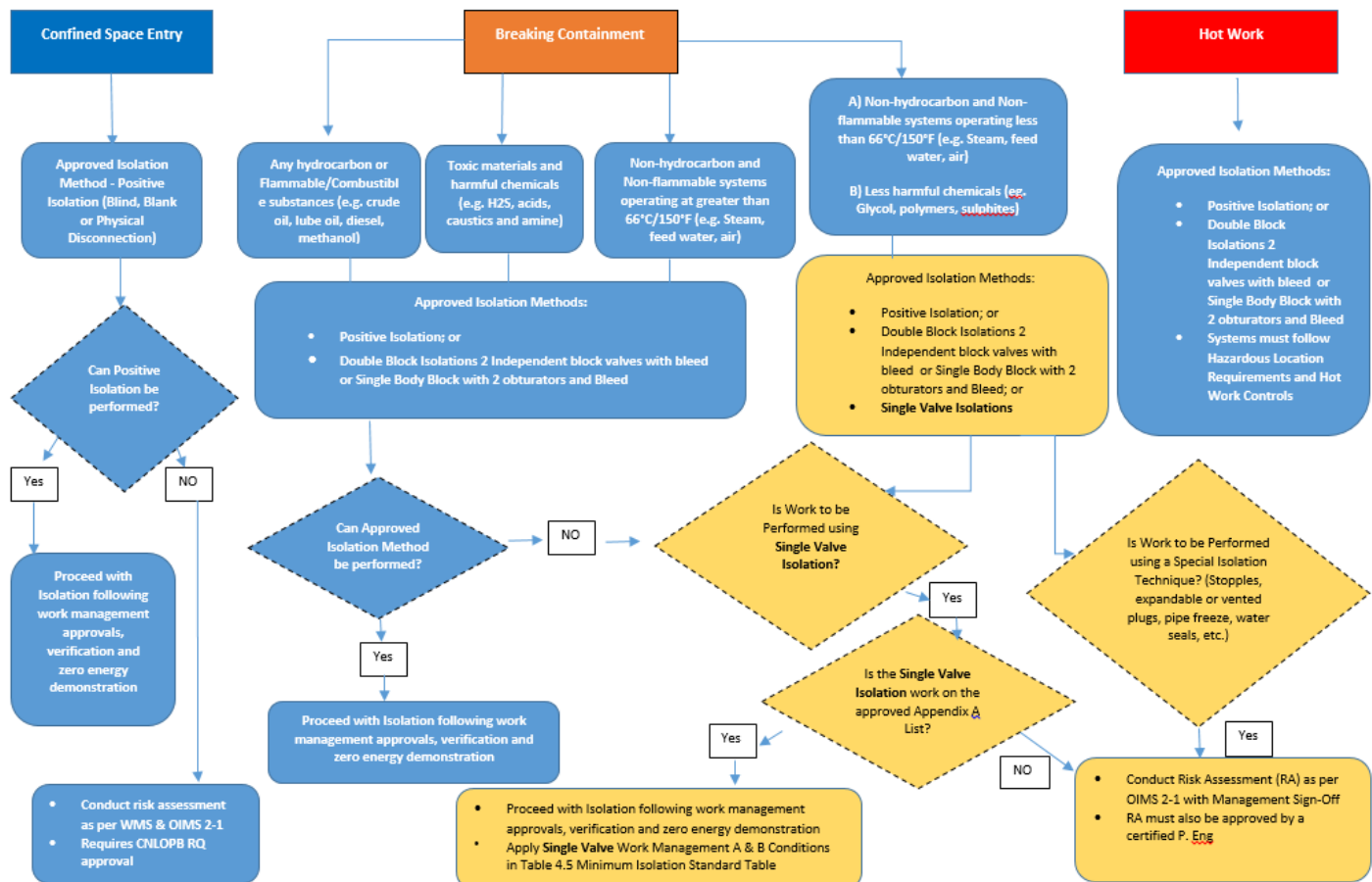
14.6 Risk Assessments

In instances where WMS requires a Risk Assessment for Piping Isolation, as per OIMS 2-1 Risk Management criteria, then the Risk Assessment will also require a Professional Engineering approval in compliance to the OHS regulations.

14.7 Piping Isolation Decision Flowchart

Piping Isolation Decision Flowchart

Scopes of Work Requiring Isolation



14.8 Hebron Approved Single Valve Isolations

PROJECT NAME: Hebron Operations EPCM Services						
CLIENT: ExxonMobil Canada Properties			CLIENT REF.: Ashley Matheson / Glen Maloney			
ORGANISATIONAL UNIT: Aker Solutions Canada Inc. MMO, St. John's			CONTRACTOR REF.:			
PROJECT NUMBER: 89000			CONTRACT NUMBER: A2507303			
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Equipment	System Description	Fluid being Isolated	Pressure Range	Temperature Range (°C) NOTE 5	Size Range	SVI Acceptable Valve Type	Zero Energy Proving Method	Comments
413	MONITOR WATER FOG DELUGE	Seawater	≤ 300#	33	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
414	FOAM SYSTEMS	Firefighting Foam	≤ 300#	33	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1
415	FIXED FIRE FIGHTING	Seawater	≤ 300#	33	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
552	GAS INJECTION WELLHEAD	Gas / Hydraulics	≤ 150# Tubing All Ranges	120 40	≤ 10"	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. If requirements are not met for flare, the individual isolation point must be evaluated prior to SVI acceptance. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4 Single Valve Acceptance on: HPU pump(s) Suction; Hydraulic Return Lines; Sight Glass on Hydraulic Tank; PHPU552154 pump and filters
559	WATER ALTERNATING GAS	Water/Gas	≤ 150#	130 (G) 87 (W)	≤ 10"	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. If requirements are not met for flare, the individual isolation point must be evaluated prior to SVI acceptance. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4
561	SURFACE FLOWLINES	Oil/Gas	≤ 150#	84	≤ 10"	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. If requirements are not met for flare, the individual isolation point must be evaluated prior to SVI acceptance. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4
565	SURFACE WELL CONTROLS	Hydraulic Fluid	≤ 150# Tubing All Ranges	30	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTES 1, 3 Single Valve Acceptance on: Hydraulic Return lines; Pilot Supply; on outlet lines from the WHCP once lines are depressured
612	HP FLARE SYSTEM	Gas	≤ 300#	200	ALL SIZES	gate, ball, globe, or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment.	NOTE 1, 4 300# lines are all ≤10"

							Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	
622	TEST SEPARATION	Oil / Gas	≤ 300#	104	ALL SIZES	gate, ball, globe, or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4 300# lines are all ≤10"

623	PRIMARY SEPARATION	Oil / Gas	≤ 300#	120	ALL SIZES	globe, ball, positive isolation spec blinds	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4 300# lines are all ≤10"
624	PRIMARY COALESCERS	Oil	≤ 150#	120	ALL SIZES	globe, ball, positive isolation spec blinds	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4
633	CRUDE STORAGE	Oil	≤ 150#	85	ALL SIZES	gate, ball, globe, or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4 N2 foam blanket implemented during caisson isolation/pump maintenance.
641	CRUDE SHIPPING OLS	Oil	≤ 300#	68	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4 300# lines are all ≤10"
650	GAS COMPRESSION	Nitrogen/Lube Oil/Diesel	≤ 300#	63	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 300# lines are all ≤10"
651	HP Gas Compressor	Gas / Condensate	≤ 300#	180	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4 300# lines are all ≤10"
657	LP GAS COMPRESSION	Gas / Condensate	≤ 300#	90	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4 300# lines are all ≤10"
659	MP GAS COMPRESSION	Gas / Condensate	≤ 300#	170	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4
673	SCALE PARAFFIN INHIBITOR	Scale Inhibitor	≤ 150#	55	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 7 SBV on returns to chem tanks (PSV's and Drains) and blanket gas N2 - DBV req for discharge Chemicals deemed toxic, but not immediately dangerous to life. In addition to zero energy confirmation, SDS PPE

								recommendations and associated WMS PPE procedures to be followed.
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Equipment	System Description	Fluid being Isolated	Pressure Range	Temperature Range (°C) NOTE 5	Size Range	SVI Acceptable Valve Type	Zero Energy Proving Method	Comments
674	DEMULSIFIERS ETC	Demulsifier	≤ 150#	55	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 7 SBV on returns to chem tanks (PSV's and Drains) and blanket gas N2 - DBV req for discharge Chemicals deemed toxic, but not immediately dangerous to life. In addition to zero energy confirmation, SDS PPE recommendations and associated WMS PPE procedures to be followed.
675	CHLORINATING SYSTEM	Biocide / O2 Scav / Defoamer	≤ 150#	55	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 7 SBV on returns to chem tanks (PSV's and Drains) and blanket gas N2 - DBV req for suction and discharge Chemicals deemed toxic, but not immediately dangerous to life. In addition to zero energy confirmation, SDS PPE recommendations and associated WMS PPE procedures to be followed.
681	PRODUCED WATER TREATMENT	Produced Water/Gas/	≤ 300#	200	ALL SIZES	gate, ball, globe, or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4
682	SEA WATER TREATMENT	NAOCl in Seawater	≤ 300#	33	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1
684	WATER DISPOSAL	Seawater	≤ 300#	58	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 Short term isolation on lines <=10" and <=300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
685	SEAWATER HANDLING SYSTEM	Seawater	≤ 300#	33	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 Short term isolation on lines <=10" and <=300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
833	MAIN POWER GENERATION	Lube Oil / Diesel	≤ 300#	80	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTES 1, 4
911	Diesel	Diesel	≤ 300#	55	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment.	NOTE 1

							Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	
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912	AVIATION FUEL STORAGE	Aviation Fuel	≤ 300#	55	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1
927	GLYCOL BASED COOLING	Glycol / Water	≤ 300#	85	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTES 1, 4
933	Hydraulic Fluid	Hydraulic Fluid	Tubing All Ranges	40	≤ 2"	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1
942	HOT WATER GLYCOL	Glycol / Water	≤ 300#	208	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTES 1, 4
951	Utility Air	Air	≤ 300#	56	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTES 1, 2 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
952	INSTRUMENT AIR DISTRIBUTION	Air	≤ 300#	56	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTES 1, 2 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
955	Air compressors	Air	≤ 300#	56	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
956	AIR DRYERS FILTERS	Air	≤ 300#	56	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
961	NITROGEN GENERATION	Nitrogen	≤ 300#	63	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.

975	SEWAGE TREATMENT	Black Water	≤ 300#	65	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment.	NOTE 1 Isolations only available around macerator. HSSE guidelines for handling blackwater to be followed.
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							Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	
976	FRESH AND UTILITY WATER	Water	≤ 300#	33	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
977	POTABLE WATER SYSTEMS	Water	≤ 300#	55	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTES 1, 2 Short term isolation on lines ≤10" and ≤300# allow Type A isolation (including butterfly). All other cases require positive isolation confirmation.
991	CLOSED DRAIN	Oil / Gas	≤ 150#	120	ALL SIZES	gate, ball, globe, or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1, 4
992	OPEN DRAIN NON-HAZARDOUS	Oily Water	≤ 300#	65	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1
999	OPEN DRAIN HAZARDOUS	Oily Water	≤ 300#	65	ALL SIZES	gate, ball, globe, butterfly (NOTE 6), or plug	Drain / Vent between isolation valve and break point to prove isolation effectiveness. Monitor pressure to ensure no pressure build-up prior to breaking containment. Isolation follows CAHE-ED-PBDES-10-600-0001 Section 12.1	NOTE 1

Hebron NOTES

NOTE 1

- Isolation aligns with Hebron Project Isolation Philosophy CAHE-ED-PBDES-10-600-0001 for isolation from a pressurized system for short-term onstream maintenance.

NOTE 2

- Control valve isolation requires appropriate drains, vents, and blinds installed between each block valve and control valve for all classes per CAHE-ED-PBDES10-600-0001 Section 12.2.1.

NOTE 3

- Pilot supply downstream of PSV5652XX-03/04 may be isolated by SVI (Type A). Upstream
- operating pressure is 5 kPSIG (34,474 kPag) and design pressure is 5.5 kPSIG (37,921 kPag) and requires DIBV (Type C) isolation. Reference vendor & IDs CAHE-PO-00003020-23-565-00C103-VDPID-0003/0004.

NOTE 4

- Pipe contents shall be cooled to $\leq 66^{\circ}\text{C}$ prior to breaking containment to minimize risk to Personnel

NOTE 5

- Temperature range listed is the system design temperature.

NOTE 6

- Single seated butterfly valves do not qualify for Single Valve Isolation (SVI) and shall not be used.

NOTE 7

- SDS and WMS Personal Protective Equipment Requirements to be followed

14.9 Hibernia Approved Single Valve Isolations



CLIENT : Hibernia Management and Development Company Ltd. : Single
PROJECT TITLE : Valve Isolation (SVI) / Appendix A Review
JOB NO. : 021736
DOCUMENT NO. : HS-O-O-X-A00-XX-0604.007 REV D14, SECTION 14.9
DOCUMENT TITLE : Work Management System - Appendix A

PROVINCE OF NEWFOUNDLAND AND LABRADOR



PERMIT HOLDER
THIS PERMIT ALLOWS

WOOD GROUP CANADA, INC.

To practice Professional Engineering
in Newfoundland and Labrador,
Permit No. as issued by PEG D0256
Which is valid for the year 2024
MIRC Signature/Number 04007

ENGINEERING STAMP



**NOTE: SPECIFIC TO HIBERNIA
INSTALLATION, SECTION 14.9 ONLY.**

REVISION RECORD							
REV.	DESCRIPTION	ORIGINATED BY		CHECKED BY		APPROVED BY	
						ENG.	PROJECTS
C1	Reissued for Use	Brandon Bursev	Shawn Jones	Shawn Jones			
		<i>BB</i> 11-Dec-24	<i>SN</i> 11-Dec-24	<i>SN</i> 11-Dec-24			
C0	Issued for Use	Ian Smith	Shawn Jones	Shawn Jones			
		IAS 14-Oct-22	SPJ 14-Oct-22	SPJ 14-Oct-22			

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Single Valve Isolation Approved List (Hibernia)

Equipment/System Description	Fluid being Isolated	Pressure Range	Temperature Range	Size Range	SVI Acceptable Valve Type	Zero Energy Proving Method	Comments
SYS 31 - CRUDE SEPARATION	HYDROCARBON LIQUID HYDROCARBON VAPOR	≤ 600# ≤ 150#	100	ALL SIZES ≤ 18"	BALL, GATE, BUTTERFLY (NOTE 2)	DRAIN, PURGE AND VENT VALVES. LOCAL INSTRUMENTATION (PI/PG)	600# EQUIPMENT LIMITED TO INSTRUMENTATION (VALVE SIZE ≤ 3"). 150# EQUIPMENT CONSISTS OF STRAINERS, COOLERS, HEATERS, BURST DISCS, COALESCERS, PSVs
SYS 31, 33, 43 - WELL ANNULI	DRILLING FLUIDS HYDROCARBON LIQUID HYDROCARBON VAPOR	≤ 5000#	200	≤ 3"	BALL, GATE	BLEED PORT	ANNULI CONSIDERED TO BE CLOSED CONTAINERS. BEFORE CONSIDERING SVI, PRESSURE MUST BE MONITORED FOR 0 INCREASE TO CONFIRM NO COMMUNICAION WITH THE RESERVIOR OR OTHER ANNULI. ZERO ENERGY MUST BE ABLE TO BE DEMONSTRATED AND CONFIRMED
SYS 33 - GAS COMPRESSORS / INJECTION	LUBE OIL SEAL OIL		200	ALL SIZES ≤ 6"	BALL, GATE	DRAIN AND BLEED VALVES, LOCAL INSTRUMENTATION (PI/PIT)	INCLUDES 1ST STAGE TURBINE AND COMPRESSOR LO SYSTEMS AND COMPRESSOR SO SYSTEM
SYS 34 - HP FLARE	HYDROCARBON VAPOR	≤ 150#	150	ALL SIZES ≤ 4"	BALL	DRAIN VALVE, LOCAL INSTRUMENTATION (PIT)	ISOLATION AGAINST GAS LIFT FLOWLINES
SYS 42 - PRODUCED WATER TREATING	PRODUCED WATER / RECOVERED OIL	≤ 300# ≤ 150#	100	ALL SIZES ≤ 14"	BALL, GATE	LOCAL INSTRUMENTATION (PI/PIT), SAMPLE CONNECTIONS, BLEED PORTS	INCLUDES STRAINERS, PUMPS, HYDROCYCLONES, INSTRUMENTATION
SYS 43 - WATER INJECTION SYS 44 - SERVICE WATER SYS 45 - FIREWATER SUPPLY SYS 46 - FRESH WATER SYS 47 - BALLAST WATER SYS 48 - POTABLE WATER SYS 50 - FIRE WATER SUPPLY (DELUGE AND SPRINKLER DISTRIBUTION)	WATER	≤ 300# ≤ 150#	45	ALL SIZES ≤ 20"	BALL, GATE, BUTTERFLY (NOTE 2)	LOCAL INSTRUMENTATION (PI/PIT), BLEED PORTS	INCLUDES FILTERS, STRAINERS, PUMPS, HYDRANTS, SPRINKLERS, INSTRUMENTATION
SYS 51 - OPEN DRAINS (NON-HAZ) SYS 52 - OPEN DRAINS (HAZ) SYS 56 - OILY WATER TREATING	PRODUCED WATER HYDROCARBON LIQUID	≤ 150#	25	ALL SIZES ≤ 4"	BALL, GATE	LOCAL INSTRUMENTATION (PI/PIT/PDIT), VENT AND BLEED PORTS	INCLUDES FILTERS, PUMPS, TANKS, CENTRIFUGE
SYS 61 - DIESEL FUEL	DIESEL	≤ 150#	30	ALL SIZES ≤ 6"	BALL, GATE	DRAIN, BLEED AND VENT VALVES. LOCAL INSTRUMENTATION (PI/PIT/PDI)	INCLUDES PUMPS, FILTERS, COALESCERS, TANKS, CENTRIFUGES, INSTRUMENTATION
	DIESEL (TURBINE)	≤ 600#	120	ALL SIZES ≤ 1.5"	GATE	DRAIN AND BLEED VALVES	FILTERS
Sys 62 - Fuel Gas	HYDROCARBON VAPOR	≤ 600# ≤ 150#	100	All Sizes ≤ 3"	BALL, GATE	DRAIN AND BLEED VALVES	600# EQUIPMENT LIMITED TO INSTRUMENTATION, AND PSVs (VALVE SIZE ≤ 3"). 300# EQUIPMENT LIMITED TO FILTERS (VALVE SIZE ≤ 6") All 150# EQUIPMENT (VALVE SIZE ≤ 6")
SYS 63 - HEATING MEDIUM	HEATING MEDIUM	≤ 150#	200	ALL SIZES ≤ 18"	BUTTERFLY (NOTE 2), GATE	DRAIN, BLEED AND VENT VALVES. LOCAL INSTRUMENTATION (PI/PIT)	INCLUDES TANKS, SUMPS, PUMPS, HEATING COILS AND DRUMS

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SYS 64 - COOLING MEDIUM	COOLING MEDIUM	≤ 150#	110	ALL SIZES ≤ 18"	BALL, GATE, BUTTERFLY (NOTE 2)	VENT AND DRAIN VALVES, LOCAL INSTRUMENTATION (PI/PIT)	INCLUDES STRAINERS, TANKS, PUMPS, COOLERS
SYS 65 - HYDRAULIC POWER	HYDRAULIC OIL		45	ALL SIZES ≤ 2"	BALL	DRAIN AND BLEED VALVES	INCLUDES PUMPS, FILTERS, ACCUMULATORS, PSVs. WELLHEAD CONTROL PANELS AND MODULES ALSO CONSIDERED UNDER SYS 65.
SYS 66 - INSTR AIR / PLANT AIR / BLACK START AIR	AIR	≤ 150#	45	ALL SIZES ≤ 4"	BALL, GATE (NOTE 3)	LOCAL INSTRUMENTATION (PI), VENT TO ATM	INCLUDES DRYERS, COMPRESSORS
SYS 67 - CHEMICAL INJECTION	WATER CLARIFIER ACID WASH ASPHALTENE INHIBITOR SCALE INHIBITOR DMO ANTIFOAM CORROSION INHIBITOR IRON SULFIDE CHELATOR BIOCIDE OXYGEN SCAVENGER H2S SCAVENGER HYPOCHLORITE SOLUTION DILUTED H2	≤ 150#	65	ALL SIZES ≤ 3"	BALL	LOCAL INSTRUMENTATION (PI, PIT), DRAIN VALVES	SBV's ON PUMP SUCTIONS. INSTRUMENTATION ON HYDROGEN RELEASE TANK
SYS 71 - MAIN POWER GENERATION	LUBE OIL	NOTE 4	120	ALL SIZES ≤ 1"	GATE	DRAIN AND VENT VALVES	LUBE OIL FILTERS
SYS 72 - ESSENTIAL POWER GENERATION	RADIATOR WATER	≤ 150#	102	ALL SIZES ≤ 2"	GATE	DRAIN VALVE	RADIATORS
SYS 83 - AFFF	AFFF	≤ 150#	20	ALL SIZES ≤ 4"	BALL	BLEED PORT	INSTRUMENTATION
SYS 89 - NITROGEN GENERATION	NITROGEN	≤ 150#	65	ALL SIZES ≤ 2"	BALL, GATE	DRAIN, BLEED AND VENT VALVES. LOCAL INSTRUMENTATION (PI/PIT)	INCLUDES ENTIRE SYSTEM

NOTES:

1. ALL SVI SHALL HAVE APPROPRIATE DRAINS, VENTS OR BLEEDS FOR PRESSURE MONITORING AND ZERO ENERGY DEMONSTRATION.
2. BUTTERFLY ARE OF HIGH PERFORMANCE DOUBLE OR TRIPLE OFFSET TYPE.
3. INCLUDES AN ACTUATED THREE WAY VALVE
4. FILTERS ARE LIMITED TO LO PUMP PSV PRESSURE OF 680 KPAG.
5. DRAINING, FLUSHING AND PURGING COMPLETED PRIOR TO BREAKING CONTAINMENT.

15.0 CSE Atmospheric Testing Reference Values and RPE Requirements

Order of Gas Testing: 1. Oxygen - Ventis Pro/MX6 2. LEL Flammable/combustible gases are tested next to determine flammability potential using- Ventis Pro/MX6. 3. Total VOC (Ultra-Rae with 10.6 eV or MiniRae 3000 with 10.6 eV lamp - if zero ppm VOC no need to test for benzene. 4. Tests for toxic gases/vapors such as benzene/CO and H ₂ S are performed last – Benzene testing using UltraRae 9.8 eV lamp with benzene sep tube: Draw Time 90 secs for summer and 211 secs for winter. Test for CO/H ₂ S using Ventis Pro. 5. For any CSE at IDLH values please consult WMS.	Class 1 CSE	Class 2 CSE	Class 2 CSE	Class 2 CSE
	Entry Allowed <u>WITHOUT</u> Respiratory Protection	½ Face RPE	Full Face RPE	Supplied Air
	Entry Allowed <u>WITH</u> Respiratory Protection ½ mask APR with OV+AG+P100 combo	Entry Allowed <u>WITH</u> Respiratory Protection Full Face APR with OV+AG+P100 combo	Entry Allowed <u>WITH</u> Respiratory Protection Full Face APR with OV+AG+P100 combo	Entry Allowed <u>WITH</u> Respiratory Protection Supplied Air
	Atmospheric Concentration			
OXYGEN based on NL Regulations	≥ 19.5% to ≤ 22.5%	< 19.5 Entry Not permitted if > 22.5%		
FLAMMABILITY: Non-Hot Work based on USM	NA Based on LEL ignition potential only *air purifying cartridges or filters are NOT effective for oxygen deficient and flammable atmospheres			> 10% of LEL (HC) Supplied Air
FLAMMABILITY: Hot Work based on USM	0% LEL (Based on LEL ignition potential only)			
TOTAL VOLATILE ORGANIC COMPOUNDS 10.6 eV Lamp	< 100 ppm	> 100 to < 1000 ppm	< 1000 ppm Full face optional – ½ face APR is protective	≥ 1000 ppm (IDLH) or unknown
Benzene	< 0.02 ppm	> 0.02 ppm - < 0.2 ppm	> 0.2 ppm - < 1 ppm	> 1 ppm to < 500 ppm (IDLH)
H ₂ S (Hydrogen sulphide)	< 1 ppm	> 1 to < 100 ppm (IDLH)		
SO ₂ (Sulphur dioxide) STEL	< 0.25 ppm	> 0.25 - < 2.5 ppm + chemical goggles	> 2.5 - < 12.5 ppm	> 12.5 to < 100 ppm (IDLH)
CO ₂ (Carbon dioxide)	< 5000 ppm	> 5000 < 40,000 ppm (IDLH)		
CO (Carbon monoxide)	< 25 ppm	> 25 to < 1200 ppm (IDLH)		
NORM (alpha/beta) (CPM = Counts Per Minute)	< 200 cpm	≥ 200 but < 1500 cpm	≥ 1500 cpm	Dose Rate Measurement (gamma)
NORM Dose Rate (gamma)	< 0.5 µSv/hr: Used in verifying/ establishing a perimeter from NORM work areas and storage areas. Measurement at the perimeter must be less than 0.5 µSv/hr. ≥ 0.5 µSv/hr: Contact Industrial Hygienist to complete dose estimates. ≥ 2.5 µSv/hr: Contact Nurse to distribute electronic dosimeters.			≥ 10 µSv/hr – Contact IH before proceeding

If levels exceed the "Requirement for no RPE" – Immediately leave the area and contact your supervisor. Consult the respiratory protection requirements

Confined Space Entry CLASS 1 – Entry without use of Respiratory Protection Equipment (RPE)

Confined Space Entry CLASS 2 – Entry where RPE is needed

WARNING: Entry into atmospheres above the class 2 limits **MUST NOT** be conducted until risk assessment has been conducted.

IMPORTANT: THE Confined Space Entry class determination table does not consider any possibility of NORM within the confined space. Presence of NORM may require the use of respiratory protection, particularly if disturbed.

NOTES:

- *All gas testing equipment is to be bump tested and/or calibrated before use*
- **Atmospheric Testing** for CSE is performed using dedicated aspirated gas monitors. **Personal** gas monitors are used as a warning of changing atmospheric conditions while inside the vessel.
- **TASK SPECIFIC RPE/PPE:** Additional information relating to PPE/RPE requirements can be found in the PPE Matrix.
- **NORM:** Bump test each day against Cs137 source
- **APR** – Air Purifying Respirator
- **OV**- Organic Vapor
- **OV+AG** – Organic Vapor & Acid Gas Combination
- **P100** – filters at least 99.97% of airborne particles and is strongly resistant to oil
- **NA** – Not Applicable

16.0 References

Work Management system Best Practice Manual - CAHE –EC-OOREF-01-006-4001-000 / HS-O-O-X-A00-XX-0604.008

Canada-Newfoundland and Labrador Offshore Area Occupational Health and Safety Regulations

17.0 Interim Approvals

Interim changes to this Appendix A must be approved by the System 6-4 Administrator and Owner.

Certificate Of Completion

Envelope Id: F1BC5676-C5B3-4C67-8208-477692E2B092

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ashley.d.matheson@exxonmobil.com

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March 9, 2025 | 18:37

ashley.d.matheson@exxonmobil.com

Security Appliance Status: Connected

Pool: Main SecApp 1

Signer Events

Ashley Matheson

ashley.d.matheson@exxonmobil.com

Production Supervisor

ExxonMobil General

Security Level: Email, Account Authentication
(None)

Signature

Signed by:

Ashley Matheson
32957A8A714D4E2...

Signature Adoption: Pre-selected Style

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Sent: March 9, 2025 | 18:39

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Signed: March 9, 2025 | 18:41

Electronic Record and Signature Disclosure:

Accepted: June 20, 2023 | 16:01

ID: c5b3660b-d14a-4a55-986f-7c06e44733d7

Company Name: Exxon Mobil Corporation

W. Duncan Fitzpatrick

w.duncan.fitzpatrick@esso.ca

President, HMDC

ExxonMobil General

Security Level: Email, Account Authentication
(None)

Signed by:

W. Duncan Fitzpatrick
7EA4532B83CB496...

Signature Adoption: Pre-selected Style

Using IP Address: 136.228.242.23

Sent: March 9, 2025 | 18:41

Viewed: March 10, 2025 | 11:08

Signed: March 11, 2025 | 06:51

Electronic Record and Signature Disclosure:

Accepted: March 21, 2023 | 12:32

ID: a562c8c1-0c6a-40ad-9bd9-cf2e1935bbdb

Company Name: Exxon Mobil Corporation

LaMont Malone

lamont.malone@exxonmobil.com

Asset Manager

ExxonMobil General

Security Level: Email, Account Authentication
(None)

DocuSigned by:

LaMont Malone
975AC8F1D5A6421...

Signature Adoption: Pre-selected Style

Using IP Address: 174.117.77.29

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Sent: March 11, 2025 | 06:51

Viewed: March 12, 2025 | 17:00

Signed: March 12, 2025 | 17:01

Electronic Record and Signature Disclosure:

Accepted: February 28, 2023 | 13:34

ID: 29660c55-71c4-4f6e-bcd9-16fcec4090f

Company Name: Exxon Mobil Corporation

Jamie Suddaby

jamie.e.suddaby@esso.ca

ExxonMobil Canada Production Manager

ExxonMobil General

Security Level: Email, Account Authentication
(None)

DocuSigned by:

Jamie Suddaby
FB60B992F204418...

Signature Adoption: Pre-selected Style

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Sent: March 12, 2025 | 17:01

Viewed: March 13, 2025 | 06:01

Signed: March 13, 2025 | 06:02

Electronic Record and Signature Disclosure:

Signer Events	Signature	Timestamp
Accepted: October 2, 2023 13:10 ID: 16a3ddac-fd8c-4e33-a100-235e25e1fec9 Company Name: Exxon Mobil Corporation		
In Person Signer Events	Signature	Timestamp
Editor Delivery Events	Status	Timestamp
Agent Delivery Events	Status	Timestamp
Intermediary Delivery Events	Status	Timestamp
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Carbon Copy Events	Status	Timestamp
Ashely Matheson ashley.d.matheson@exxonmobil.com Production Supervisor ExxonMobil General Security Level: Email, Account Authentication (None) Electronic Record and Signature Disclosure: Accepted: June 20, 2023 16:01 ID: c5b3660b-d14a-4a55-986f-7c06e44733d7 Company Name: Exxon Mobil Corporation	COPIED	Sent: March 13, 2025 06:02 Resent: March 13, 2025 06:02 Viewed: March 13, 2025 06:05
Witness Events	Signature	Timestamp
Notary Events	Signature	Timestamp
Envelope Summary Events	Status	Timestamps
Envelope Sent	Hashed/Encrypted	March 9, 2025 18:39
Certified Delivered	Security Checked	March 13, 2025 06:01
Signing Complete	Security Checked	March 13, 2025 06:02
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